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Assignment

Brain tumor Segmentation and Survival Time Prediction

Course: Deep Learning and Its Applications - CO3133

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1 Introduction

Brain tumors represent one of the most lethal classifications of disease, accounting for a significant number of fatalities globally each year. Despite rapid advancements in healthcare, the medical community continues to face substantial challenges in developing non-invasive or streamlined procedures for tumor resection and treatment. Currently, the success of such interventions relies heavily on the "human factor"—neurosurgeons must possess extraordinary manual dexterity and decades of specialized clinical experience to navigate the intricate structures of the brain without causing collateral damage.

However, the modern era of Artificial Intelligence (AI) is beginning to bridge these gaps. Machine learning and deep learning models are revolutionizing the pre-surgical pipeline by providing high-precision diagnostic support at pre-surgical process by differentiating HGG vs LGG, predicting survival time, grading gliomas[1],...

In this assignment, we leverage state-of-the-art deep learning architectures to address the complexities of brain tumor analysis. Specifically, we implement the Swin UNetR framework, which utilizes a hierarchical Vision Transformer (Swin Transformer) as the encoder to capture multi-scale contextual information for precise semantic segmentation of tumor boundaries. Furthermore, we reimplement ResDeepSurv, a deep residual network designed for survival time prediction, to integrate clinical prognosis into our pipeline.

For our data foundation, we utilize the widely recognized BraTS 2020 dataset[2]. Due to the requirements for supervised learning, our implementation focuses exclusively on the training subset. This decision was made because the training folder provides the essential "ground truth" masks (labels) necessary for model optimization, whereas the validation folder lacks these annotations, rendering it unsuitable for local training and evaluation metrics.

The performance of each model is rigorously compared using evaluation metrics tailored for Segmentation are Dice Score, 95th percentile Hausdorff Distance (HD95) and Intersection over Union (IoU). For Survival Time Prediction, metrics used are Concordance index (C-Index) and Integrated Brier score (IBS).

Here is our Github link that contain the assignment:

2 Related work

2.1 Dataset

The Multimodal Brain Tumor Segmentation (BraTS) 2020 dataset serves as the primary data foundation for this assignment, providing a standardized benchmark for evaluating state-of-the-art volumetric segmentation algorithms. Curated for the annual MICCAI challenge, the dataset comprises clinically-acquired, multi-institutional 3D MRI scans from 369 training subjects. Each case includes four distinct physical modalities: native T1-weighted (T1), post-contrast T1-weighted (T1ce), T2-weighted (T2), and Fluid Attenuated Inversion Recovery (FLAIR) volumes. These modalities are co-registered to the same anatomical template, skull-stripped, and resampled to a consistent 1 mm^3 isotropic resolution. To facilitate supervised learning, expert neuroradiologists provided voxel-level ground truth annotations, categorizing tissue into the necrotic and non-enhancing tumor core (NCR/NET), peritumoral edema (ED), and GD-enhancing tumor (ET). For the scope of this project, we utilize the training subset exclusively, as it contains the complete set of labels required for optimizing the Swin UNetR and ResDeepSurv architectures.

Figure 1: Example of brain slice sample. The fifth image is flair with mask

The original image has size $250 \times 250 \times 155$, but most of the image is black void (background), which contains zero information and may introduce noise that degrades the model's accuracy. We preprocess the data by center cropping it to $192 \times 192 \times 144$, then zooming to $128 \times 128 \times 128$ so the GPU does not have to process large images. Raw MRI intensities are arbitrary—the same tissue can have value 800 on one scanner and 2400 on another. Z-score normalization transforms the data to have mean ≈ 0 and standard deviation ≈ 1, making training stable across scanners. Background voxels are exactly 0 by convention in BraTS 2020. If included in the mean and standard deviation calculation, they would dominate the statistics (since most voxels are zero), distorting normalization of actual brain tissue. Therefore, background remains 0 after normalization. In addition, we augment a small portion by shifting from -0.1 to 0.1 and scaling from 0.9 to 1.1.

We split dataset into 80:20 ratio for train and test dataset respectively and in the train dataset, we use k-fold for cross validation (K = 5).

2.2 Radiomic Feature Extraction and Engineering Pipeline

To transform multi-modal MRI scans into quantitative representations suitable for survival analysis, patient-level radiomic features were systematically extracted using the open-source PyRadiomics library. The extraction process leveraged the FLAIR and T1ce modalities in conjunction with their corresponding 3D segmentation masks, including Whole Tumor (WT), Tumor Core (TC), and Enhancing Tumor (ET). This process yielded a comprehensive set of features,

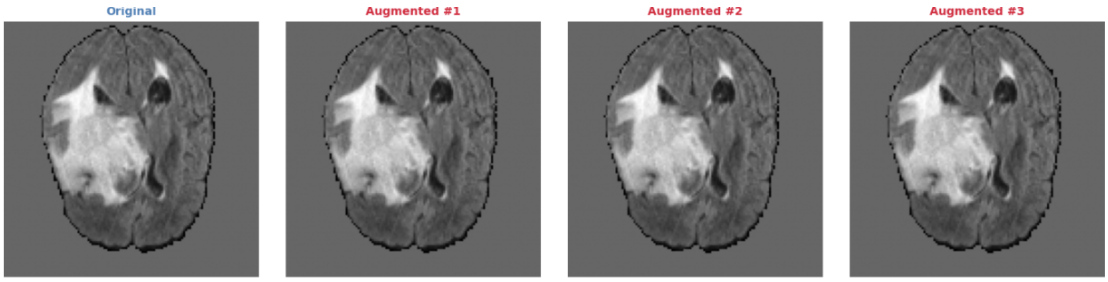

Figure 2: Example of data augmentation

encompassing shape descriptors, first-order intensity statistics, and higher-order texture features. Additionally, Laplacian of Gaussian (LoG) filtering was applied to capture fine-grained microenvironmental patterns.

To further enhance the representational capacity, the radiomic feature set was augmented with engineered volumetric ratios (TC/WT and ET/WT), as well as key clinical variables, including patient age and one-hot encoded Extent of Resection (EOR). This integration enabled the construction of a holistic feature space that combines imaging-derived biomarkers with clinically relevant information.

Given the high dimensionality of the extracted features, a rigorous feature engineering pipeline was implemented to mitigate the curse of dimensionality while strictly preventing data leakage. The dataset was first partitioned into an 80% training set and a 20% held-out test set. All preprocessing steps were exclusively fitted on the training data and subsequently applied to the test set.

The radiomic features underwent median imputation to handle missing values, followed by a collinearity reduction step, where features exhibiting a Pearson correlation coefficient greater than 0.90 were removed to eliminate redundancy. The remaining features were normalized using a robust scaling strategy to reduce the influence of outliers. Subsequently, a univariate feature selection method based on the ANOVA F-statistic was applied to retain the top 50 most prognostic features.

The final outcome of this pipeline is a compact and optimized feature representation, which serves as the input for both the proposed ResDeepSurv model and the XGBoost baseline. This design ensures a fair and consistent comparison while maintaining strong generalization capability.

2.3 Loss function

In this study, we address the multi-task nature of the BraTS 2020 challenge by employing two distinct loss functions tailored to the specific objectives of each task. For the segmentation task, the loss function is designed to handle class imbalance and promote accurate delineation of tumor subregions. In parallel, for the auxiliary task, a complementary loss is adopted to optimize its respective predictive objective. The joint use of these loss functions enables effective multi-task learning, facilitating improved overall model performance through shared representations and task-specific optimization.

2.3.1 Soft Dice loss

In this work, we employ the soft Dice loss as the primary objective for the segmentation task, owing to its robustness in handling class imbalance in medical image analysis. The loss is computed based on the overlap between predicted probability maps and ground truth labels. To ensure numerical stability, particularly in the presence of small target regions, smoothing terms are incorporated into both the numerator and denominator, with $\text{smooth}_{nr} = 1 \times 10^{-5}$ and $\text{smooth}_{dr} = 1 \times 10^{-5}$.

Crucially, due to the nested structure of tumor sub-regions in the BraTS dataset—where a voxel may simultaneously belong to Whole Tumor (WT), Tumor Core (TC), and Enhancing Tumor (ET)—a sigmoid activation function is applied to the network outputs (sigmoid = true). This formulation produces independent probability maps for each class prior to loss computation. Such a design enables effective modeling of overlapping regions, stabilizes training, and promotes accurate segmentation by directly maximizing the overlap between predictions and ground truth across all channels.

We use the soft Dice loss function, which is computed in a voxel-wise manner as For class c:

$$L_c = 1 - \frac{2\sum_{i=1}^{N} p_{i,c} g_{i,c} + \text{smooth}_{nr}}{\sum_{i=1}^{N} p_{i,c} + \sum_{i=1}^{N} g_{i,c} + \text{smooth}_{dr}} \tag{1}$$

where $p_{i,c}$ is the predicted probability (after sigmoid) for voxel i in class c, $g_{i,c}$ is the ground-truth value (0 or 1) for voxel i in class c, N is the total number of voxels, and smooth_{nr} and smooth_{dr} are smoothing constants added to avoid division by zero.

2.3.2 Cox Loss

In this study, the XGBoost model is not trained using conventional regression objectives such as Mean Squared Error (MSE) to predict absolute survival time. Instead, we employ the Negative Cox Partial Likelihood loss (Cox loss) to directly optimize the ranking of patient survival risk.

The Cox loss formulation can be interpreted as a multi-class classification problem at each observed event time. Let $R(y_i)$ denote the *risk set*, which consists of all individuals j such that $y_j \geq y_i$. At time y_i, the conditional probability that individual i experiences the event is defined as:

$$P_i = \frac{\exp(f(x_i))}{\sum_{j \in R(y_i)} \exp(f(x_j))}, \tag{2}$$

where $f(x)$ represents the predicted risk score produced by the XGBoost model.

The objective function is derived by minimizing the negative log-partial likelihood:

$$\mathcal{L}_{Cox} = -\sum_{i:E_i=1} \log P_i = -\sum_{i:E_i=1} \left(f(x_i) - \log \sum_{j \in R(y_i)} \exp(f(x_j)) \right). \tag{3}$$

By minimizing this loss, the gradient boosting process iteratively adjusts the model to assign higher risk scores to patients with shorter survival times. This ensures that the predicted risk ordering is consistent with the observed survival outcomes.

The use of Cox loss is particularly suitable for medical data, where preserving the correct ranking of patient risk is often more clinically meaningful than predicting exact survival durations.

2.4 Evaluation Metrics

Segmentation Metrics. To comprehensively assess the segmentation performance, we employ a set of widely used evaluation metrics, including **Dice Score**, **Intersection over Union (IoU)**, **95% Hausdorff Distance (HD95)**, and **Sensitivity**. The **Dice Score** and **IoU** measure the degree of overlap between the predicted segmentation and the ground truth, providing complementary insights into region-wise accuracy.

The **HD95** evaluates boundary agreement by computing the 95th percentile of the Hausdorff distance, thereby reducing sensitivity to outliers while capturing spatial discrepancies between contours. Let P represent the predicted point set and G represent the ground truth point set. The directed Hausdorff distance from P to G is defined as:

$$h(P, G) = \max_{p \in P} \min_{g \in G} \|p - g\|_2. \tag{4}$$

The bidirectional Hausdorff Distance (HD) is then given by:

$$HD(P, G) = \max(h(P, G), h(G, P)). \tag{5}$$

The **HD95** is computed by replacing the maximum operator with the 95th percentile of the distances, ensuring robustness against noise and irregular boundaries.

Sensitivity, also known as recall, quantifies the model's ability to correctly identify positive voxels, which is particularly important in medical imaging tasks where missing pathological regions can have critical consequences. Together, these metrics provide a comprehensive evaluation of both volumetric overlap and boundary precision.

Survival Prediction Metrics. For the survival prediction task, we evaluate model performance using the **Concordance Index (C-index)** and the **Integrated Brier Score (IBS)**.

The **C-index** measures the model's ability to correctly rank patients based on their predicted risk scores, reflecting how well the predicted ordering aligns with the observed survival times. It is defined as:

$$C = \frac{1}{n} \sum_{\substack{i=1 \\ e_i=1}}^{n} \sum_{\substack{j=1 \\ y_i<y_j}}^{n} \mathbb{I}(yr_i > yr_j) \tag{6}$$

where n is the number of samples, e_i is the event indicator for sample i, y_i denotes the observed survival time, yr_i represents the predicted risk score, and $\mathbb{I}(\cdot)$ is the indicator function.

In contrast, the **IBS** evaluates the overall accuracy of predicted survival probabilities over time by integrating the Brier Score over a time interval $[0, t_{\max}]$. The Brier Score at time t is formulated using the Inverse Probability of Censoring Weighting (IPCW) approach as follows:

$$BS(t) = \frac{1}{n} \sum_{i=1}^{n} \left\{ \frac{\left[0 - \hat{S}(t \mid x_i)\right]^2 \mathbb{I}(y_i \leq t,\, e_i = 1)}{\hat{G}(y_i)} + \frac{\left[0 - \hat{S}(t \mid x_i)\right]^2 \mathbb{I}(y_i > t)}{\hat{G}(t)} \right\}. \tag{7}$$

Here, n denotes the sample size, $\mathbb{I}(\cdot)$ is the indicator function, y_i is the observed survival time, and e_i is the event indicator (with $e_i = 1$ indicating the occurrence of the event). The function $\hat{G}(t)$ represents the Kaplan–Meier estimate of the censoring distribution, while $\hat{S}(t \mid x_i)$ denotes the estimated survival probability for subject i given covariates x_i. The true survival status is defined as 0 if the event occurs before time t, and 1 if the subject survives beyond time t.

The Integrated Brier Score (IBS) is then defined as:

$$IBS = \frac{1}{max(y_i)} \int_0^{max(y_i)} BS(t)\, dt, \tag{8}$$

The **IBS** captures both calibration and discrimination aspects of the model, with lower values indicating superior performance. Together, these metrics provide a comprehensive evaluation of the model's capability in survival prediction, accounting for both ranking accuracy and probabilistic prediction quality in the presence of censored data.

3 Proposed method

3.1 Project overview

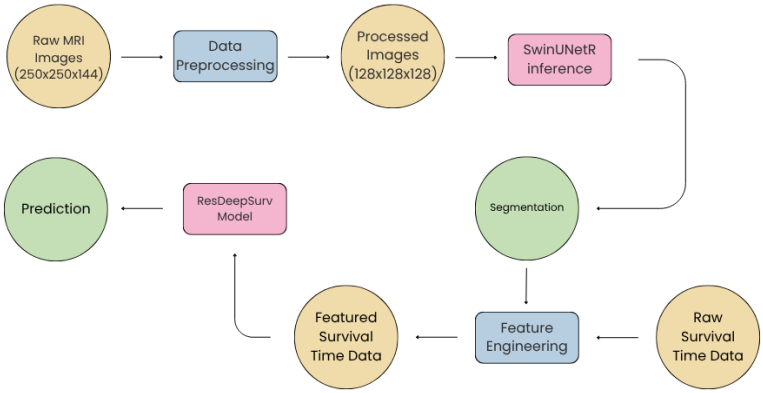

Figure 3: Our pipeline to handle segmentation and survival time prediction

Figure 3 presents the full methodology used in this project, linking image segmentation and survival modeling in a single, clinically meaningful workflow. The process starts from four co-registered MRI modalities (T1, T1ce, T2, and FLAIR), which are first standardized through the preprocessing strategy described in the previous subsection. This step ensures that all inputs are spatially aligned, intensity-normalized, and computationally manageable, so downstream models can focus on disease-relevant structure rather than scanner-specific variation. After preprocessing, the multimodal volume is passed to Swin UNETR, which serves as the core segmentation backbone. Unlike purely convolutional encoders, Swin UNETR combines local feature extraction with hierarchical self-attention, allowing the network to capture both fine tumor boundaries and broader anatomical context across 3D space. This is particularly important in glioma analysis, where edema can be diffuse while enhancing regions are compact and irregular. The output of the first branch is a voxel-wise segmentation map for the major tumor subregions.

For survival prediction, we adopt ResDeepSurv as the second stage of the pipeline. This module extends neural survival analysis with residual connections, improving optimization stability

and enabling deeper nonlinear modeling without severe gradient degradation. The model estimates individualized risk under a Cox-style objective, so it can naturally handle right-censored observations that are common in longitudinal oncology datasets. In practice, this allows the network to learn risk ordering among patients rather than forcing a naive regression target for time-to-event. The predicted risk scores can then be transformed into survival stratification groups (e.g., lower-, intermediate-, and higher-risk cohorts), supporting downstream clinical decision support.

Finally, the overall design is scalable. As additional data become available, the same framework can incorporate improved segmentation backbones, richer handcrafted or learned imaging features, and expanded clinical variables without changing the central logic of the method. Therefore, the pipeline in Figure 3 is not only an implementation detail, but the main contribution of our study: a unified bridge from volumetric tumor characterization to personalized survival prediction.

3.2 Architecture

3.2.1 Swin-UNETR

Swin-UNETR is a U-shaped 3D medical image segmentation network that combines a hierarchical Swin Transformer encoder with a convolutional decoder. Instead of relying only on local convolution kernels, the encoder partitions the input volume into patches and applies shifted-window self-attention to capture both local details and longer-range dependencies efficiently. Multi-scale features extracted at different resolutions are then passed to the decoder through skip connections, allowing the model to recover fine spatial details while preserving rich contextual information. This design makes Swin-UNETR well suited for segmenting brain tumors with large variation in size, shape, and appearance.

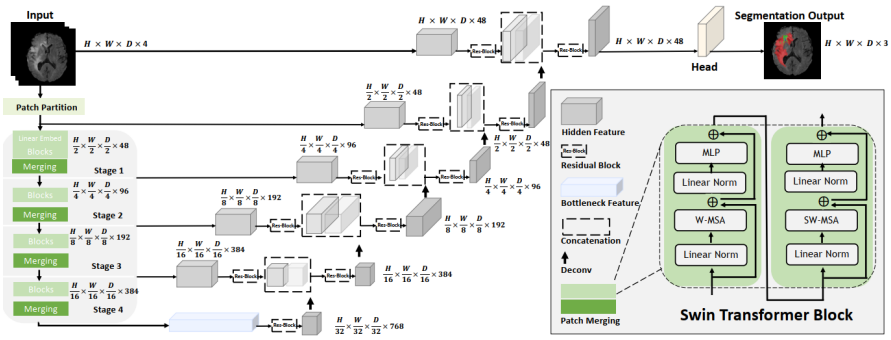

Figure 4: Swin UNETR architecture

3.2.2 ResDeepSurv

ResDeepSurv first maps the input variables into a feature representation X, which is then processed by a stack of residual units. Unlike convolution-based residual blocks, each unit is built from two linear layers, batch normalization, and a nonlinear activation function. This design is

intended to model complex nonlinear relationships among clinical and imaging features while preserving stable optimization through residual learning. The residual mapping is defined as

$$X_{l+1} = F(X_l; \{W_l, W_{l+1}\}) + X_l \tag{9}$$

where W_l and W_{l+1} are the parameters of the two linear layers, and $F(\cdot)$ denotes the learned residual transformation. By learning the residual term instead of a full direct mapping, the network can capture richer feature interactions and improve representational capacity.

To further emphasize informative variables, ResDeepSurv incorporates a self-attention mechanism that adaptively assigns weights to different features. Since some variables are more relevant than others for survival prediction, this attention layer helps the model focus on the most discriminative information and better capture dependencies among features. The attention computation is expressed as

$$A = \text{softmax}(X_I X_I^T) \tag{10}$$

$$X_O = A X_I \tag{11}$$

where X_I and X_O denote the input and output feature representations, respectively, and A is the normalized attention matrix. The *softmax* operation constrains the attention weights to the range $[0, 1]$ and ensures that they sum to one, allowing the network to learn the relative importance of each feature in a data-driven manner.

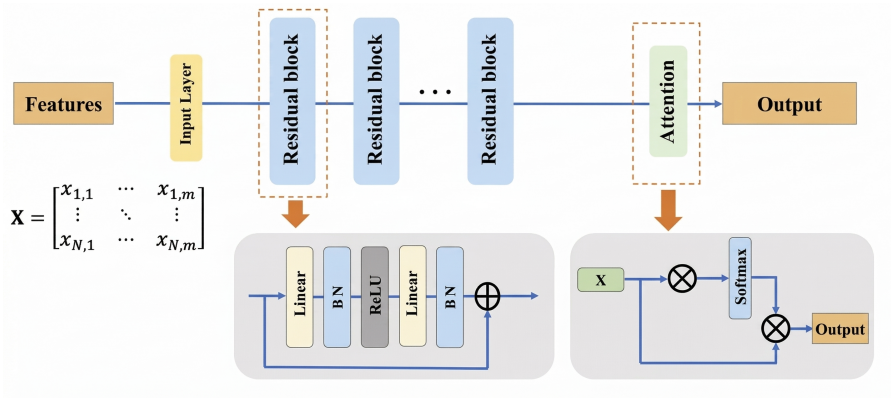

Figure 5: ResDeepSurv architecture

4 Evaluation

4.1 Segmentation using SwinUnetr

4.1.1 Training Methodology

In this study, we adopt a 5-fold cross-validation strategy to ensure robust model evaluation and to mitigate overfitting. The segmentation model is based on the SwinUNETR architecture with a hierarchical depth configuration of (2, 4, 2, 2), and gradient checkpointing is employed to reduce memory consumption during training. The model takes 4-channel 3D multi-modal MRI inputs and predicts probability maps corresponding to three tumor sub-regions.

Optimization and Learning Rate Scheduling. Model parameters are optimized using the AdamW optimizer, with an initial learning rate of 10^{-3} and a weight decay of 10^{-5}. To improve training stability, particularly during the early stages, we employ a two-phase learning rate schedule. Specifically, a linear warmup phase gradually increases the learning rate from 10^{-7} to 10^{-3}, followed by a cosine annealing schedule that decays the learning rate to a minimum of 10^{-6}.

Validation and Early Stopping. Within each fold, model performance is evaluated at the end of every training epoch. To prevent overfitting, early stopping is implemented with a patience of five epochs, based on the mean Dice score across all classes. The model checkpoint that achieves the highest validation Dice score is selected as the final model for each fold.

3D Inference Strategy. Due to the high memory demands of 3D medical image processing, we employ a sliding window inference strategy during validation and testing. Each volume is divided into overlapping patches of size $128 \times 128 \times 128$, with an overlap ratio of 70%. A Gaussian weighting scheme is applied during patch aggregation to ensure smooth transitions and to reduce boundary artifacts.

Metric Evaluation and Tracking. Model performance is quantitatively assessed using multiple spatial metrics, including the Dice coefficient, Intersection over Union (computed as $IoU = \frac{Dice}{2-Dice+10^{-8}}$), 95% Hausdorff Distance (HD95), and Sensitivity. Throughout training, all loss values, hyperparameter configurations, and evaluation metrics are systematically logged using Weights & Biases (W&B) for monitoring and reproducibility.

Final Model Selection. After completing the cross-validation process, the best-performing model is selected based on the highest validation Dice score across all folds. This optimal model is subsequently evaluated on a held-out test set to provide an unbiased estimate of its generalization performance on unseen data.

4.1.2 Performance and Analysis

4.2 Segmentation Performance and Analysis

To evaluate the real-world generalization capability of the proposed SwinUNETR model, the optimal checkpoint identified during cross-validation (Fold 4) was assessed on a strictly held-out 20% test set. Model performance was quantified using four spatial metrics and analyzed across the three clinically relevant BraTS sub-regions: Whole Tumor (WT), Tumor Core (TC), and Enhancing Tumor (ET). The final results are summarized in Table 1.

The model demonstrates strong generalization performance, achieving a global average Dice score of approximately 0.85 on unseen data. A detailed class-wise analysis reveals several important technical and clinical insights.

Table 1: Segmentation Performance on Test Set

Evaluation Metric	Whole Tumor (WT)	Tumor Core (TC)	Enhancing Tumor (ET)
Dice Score	0.9075	0.83147	0.80632
IoU	0.83066	0.67549	0.71155
Sensitivity	0.91369	0.8519	0.81695
HD95 (mm)	6.08994	6.05572	4.21212

Whole Tumor (WT): The model achieves the highest performance on the WT region, with a Dice score of 0.9075 and Sensitivity of 0.91369. This is consistent with expectations, as the WT represents the largest contiguous tumor volume and exhibits relatively clear boundaries.

Tumor Core (TC): The TC region is segmented with high stability, achieving a Dice score of 0.83147 and Sensitivity of 0.8519. The model effectively differentiates the tumor core from surrounding edema.

Enhancing Tumor (ET): The ET region remains the most challenging due to its small size and irregular structure. Although it yields the lowest Dice score (0.80632), the model achieves the lowest boundary error (HD95 = 4.21212 mm), indicating high boundary precision.

From a clinical perspective, the combination of high sensitivity and low boundary error is particularly significant. The relatively low HD95 values, especially for the ET region, suggest that the model produces spatially accurate tumor boundaries even in challenging cases.

The overall segmentation pipeline demonstrates a stable and robust optimization framework. The combination of SwinUNETR, Dice-based optimization, sliding-window inference, and early stopping ensures strong generalization and reliable performance on unseen data.

To visually assess the segmentation quality, the following examples illustrate our model's outputs alongside the ground-truth annotations, highlighting the model's spatial precision across different tumor sub-regions.

For further details regarding the training and evaluation procedures, the full experiment logs can be accessed via the following link: `https://api.wandb.ai/links/beefsteak/wv1oo63c`.

4.3 Survival Prediction using ResDeepSurv model

4.3.1 Training methodology

In this study, we also adopt a 5-fold cross-validation strategy to ensure robust model evaluation and to mitigate overfitting for the overall survival prediction task. The survival model is based on the ResDeepSurv architecture, which takes extracted patient-level features as input and predicts continuous risk scores corresponding to the relative hazard of each patient.

Optimization and Learning Rate Scheduling. Model parameters are optimized using the AdamW optimizer with a weight decay of 10^{-5}. The network is trained by minimizing the Cox Proportional Hazards (CoxPH) loss. To effectively navigate the complex survival loss landscape and ensure stable convergence, we employ a cosine annealing learning rate schedule that gradually decays the learning rate from an initial value (e.g., 8×10^{-4} or 10^{-3}) to a minimum threshold of 10^{-5} over the course of training.

Validation and Early Stopping. Within each fold, model performance is evaluated at the end of every training epoch. To prevent overfitting, early stopping is applied with a patience of 15 epochs, based on the validation Concordance Index (C-index). The model checkpoint achieving the highest validation C-index is selected as the optimal model for that fold.

Ensemble Inference Strategy. Given the high predictive variance commonly observed in

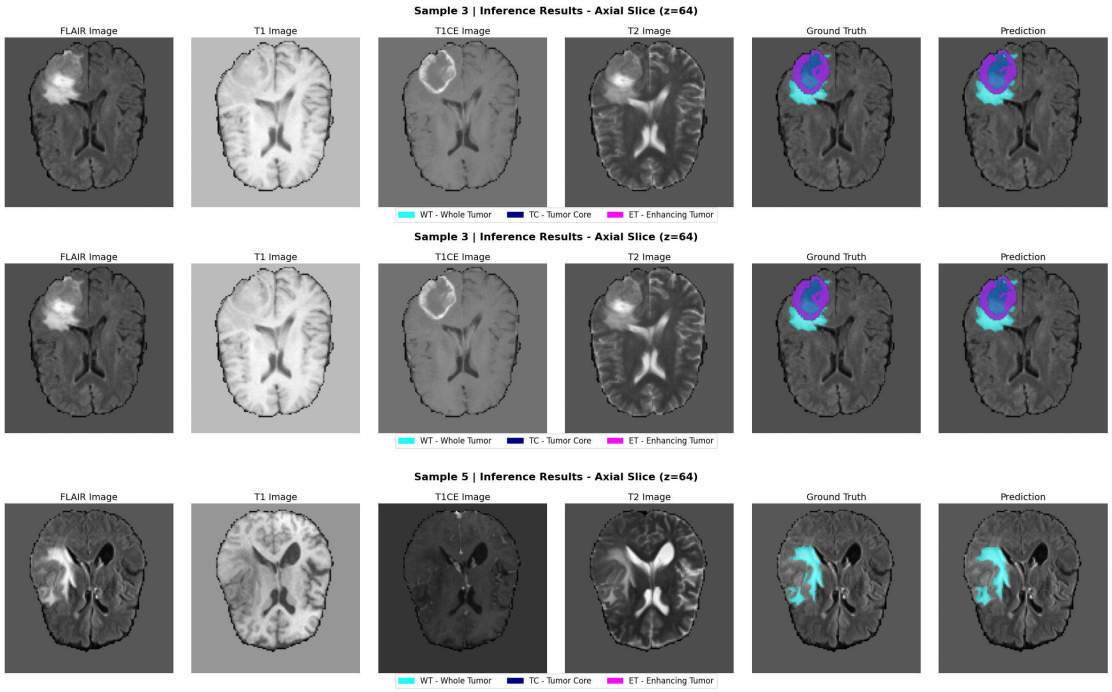

Figure 6: Qualitative segmentation results on representative test cases.

survival analysis with limited clinical datasets, we adopt an ensemble inference strategy during testing. Instead of selecting a single best-performing fold, the final predicted risk score for each patient is computed as the average of the predictions from all five cross-validation models, thereby improving robustness and stability.

Metric Evaluation and Tracking. Model performance is evaluated using standard survival analysis metrics, including the Concordance Index (C-index) to assess ranking capability and the Integrated Brier Score (IBS) to measure the accuracy of predicted survival probabilities over time. The IBS is computed using Breslow's method to estimate the baseline cumulative hazard. Throughout training, all loss values, hyperparameter configurations, and evaluation metrics are systematically logged using Weights & Biases (W&B) to ensure transparency and reproducibility.

Final Evaluation. After completing cross-validation, the ensemble model is evaluated on a strictly held-out test set. This approach provides a stable and unbiased estimate of the model's generalization performance and its effectiveness in predicting survival outcomes on unseen data.

4.3.2 Results and Analysis

To evaluate the real-world generalization capability of the proposed ResDeepSurv model, the ensemble inference strategy formulated during cross-validation was assessed on a strictly held-out 20% test set. Model performance was quantified using two standard time-to-event metrics and compared against a strong machine learning baseline (XGBoost). The final results are summarized in Table 2.

The model demonstrates competitive generalization performance on unseen data, achieving a final Concordance Index (C-index) of 0.5772. While the absolute metric values reflect the inher-

Table 2: Final test set evaluation metrics for survival prediction.

Model Paradigm (Algorithm)	**C-Index ↑**	**IBS ↓**
Machine Learning Baseline (XGBoost)	0.5735	0.1243
Deep Learning Ensemble (ResDeepSurv)	**0.5772**	**0.1155**

ent complexity of overall survival prediction in neuro-oncology, a detailed comparative analysis reveals several important technical and clinical insights.

Discriminative Performance (C-index): The ResDeepSurv ensemble achieves superior ranking performance compared to the baseline, yielding a C-index of 0.5772. Survival prediction for glioblastoma is inherently challenging due to tumor heterogeneity, high clinical noise, and substantial right-censoring. Despite these challenges, the model successfully learns latent feature representations that enable more accurate temporal risk ranking than the handcrafted features used in the XGBoost baseline (0.5735).

Calibration Accuracy (IBS): The model demonstrates stable calibration, achieving an IBS of 0.1155. This indicates that the predicted survival probabilities are reasonably consistent with observed outcomes across the follow-up period. The improvement over the baseline (0.1243) highlights the effectiveness of the ResDeepSurv architecture in modeling time-dependent survival dynamics.

Impact of Ensemble Strategy: Survival prediction from high-dimensional features is prone to high variance, particularly in small clinical datasets. While individual fold models may exhibit unstable performance, the ensemble approach effectively aggregates diverse learned representations, reducing variance and stabilizing predictions. This consensus-based mechanism enables the deep learning model to outperform the machine learning baseline.

From a clinical perspective, simultaneous improvements in both discriminative ability (C-index) and calibration (IBS) are meaningful. Even modest gains suggest more reliable and temporally consistent prognostic predictions, which are essential for patient risk stratification, personalized treatment planning, and post-operative management.

Overall, the survival prediction pipeline demonstrates a robust and well-structured optimization framework. The combination of the ResDeepSurv architecture, 5-fold cross-validation, AdamW optimization, cosine annealing learning rate scheduling, and ensemble inference effectively addresses the challenges of censored survival data. Additionally, early stopping based on validation C-index ensures strong generalization, resulting in a stable and reliable predictive model on unseen data.

For further details regarding the training and evaluation procedures, the full experiment logs can be accessed via the following link: `https://api.wandb.ai/links/beefsteak/6wfupl8k`.

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